

COMP0086 Past Exam Questions

Common Topics with COMP0171 — Full Question Extraction

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1 Topic 1: Free Energy / ELBO / EM Algorithm

1.1 2017 Q1: Free Energies [20 marks]

Also appears verbatim as 2022 Q5.

In a fit of confusion, one of your colleagues is attempting to derive the expectation maximisation (EM) algorithm for a mixture model using the free energy, but has forgotten the entropy term.

That is, given data $\{x_i\}$ and a generic mixture model with M components and parameters $\pi, \theta_1, \dots, \theta_M$:

$$s_i \sim \text{Discrete}(\pi), \quad s_i \in \{1 \dots M\}; \quad \pi_m \geq 0; \quad \sum_{m=1}^M \pi_m = 1$$

$$x_i | s_i = m \sim P_m(x_i; \theta_m), \quad x_i \in \mathbb{R}^D,$$

he has introduced posterior responsibilities $r_i^m \geq 0$ with $\sum_{m=1}^M r_i^m = 1$ for each observation i , and is seeking to maximise

$$\tilde{\mathcal{F}}(\{r_i^m\}, \{\theta_m\}, \pi) = \sum_i \sum_m r_i^m \log P(x_i, s_i = m | \theta_m, \pi),$$

with respect to $\{r_i^m\}$, $\{\theta_m\}$ and π .

However, he quickly runs into a problem. When using the method of Lagrange multipliers to find the optima with respect to r_i^m (as we did in lecture) he obtains derivatives in which the r_i^m do not appear. He calls you over to help.

- (a) You suggest thinking geometrically. Taking the case $M = 2$, sketch the isovalue contours for $\tilde{\mathcal{F}}$ as a function of the responsibilities for the first data point, as well as the constraint surface $\sum_m r_1^m = 1$. Argue that the maximum will typically be found when $r_1^m = 1$ for some m . What determines which m ? [5 marks]
- (b) Derive the correct free energy for this model, showing explicitly that it lower bounds the log-likelihood. You should start from the log-likelihood for the model above, and complete the derivation using the specific model notation — do **not** reproduce a generic derivation and substitute the model-specific values at the final step. [5 marks]
- (c) Show that the incorrect free energy $\tilde{\mathcal{F}}$ is also a lower bound on the log likelihood. [3 marks]
- (d) Now consider a continuum of free energies $\mathcal{F}_\tau$, $\tau \in [0, 1]$ in which the entropy term is premultiplied by τ . Derive the optimal responsibilities for $\mathcal{F}_\tau$ when $\tau > 0$. Then take the limit $\tau \rightarrow 0$ and show that you obtain the same result as you found geometrically. [5 marks]
- (e) What well-known algorithm will result if you maximise $\tilde{\mathcal{F}}$ for a mixture of Gaussians with fixed identity covariance matrices and unknown means, and fixed mixing proportions $\pi_m = 1/M$? [2 marks]

1.2 2019 Q3: Free-Energy Derivatives [20 marks]

Also appears verbatim as 2022 Q3.

Consider a general latent variable model with observed data $\mathcal{X}$, latent variables $\mathcal{Z}$ and parameters θ . The log-likelihood function is given by

$$\ell(\theta) = \log \int d\mathcal{Z} P(\mathcal{X}, \mathcal{Z} | \theta).$$

- (a) How is the free energy defined? Show that it is a lower bound on the log likelihood. [4 marks]

- (b) Show that the (generalised) M-step gradient evaluated after an exact E-step is (under some regularity conditions) equal to the gradient of log likelihood with respect to the parameters. [6 marks]
- (c) Consider a mixture of spherical Gaussians:

$$p(z_n) = \text{Discrete}(\pi), \quad p(x_n|z_n) = \prod_k (\mathcal{N}(\mu_k, I))^{\delta_{z_n, k}}$$

where $z_n \in 1 \dots K$ and $\delta_{z_n, k}$ is an indicator taking the value 1 if $z_n = k$ and 0 otherwise.

Compute the exact M-step update for the parameter μ_k , as well as the gradient, and show that the exact step changes μ_k along the direction of the gradient. [6 marks]

- (d) Is this generally true? That is, is the change in parameters resulting from an exact (i.e. non-generalised) M-step always aligned with the gradient direction? Explain. [4 marks]

2 Topic 2: Conjugate Priors / Bayesian Linear Regression

2.1 2018 Q1: Conjugate Mines [20 marks]

Also appears verbatim as 2020 Q1.

- (a) Define a conjugate prior. [2 marks]
- (b) How are the parameters of the posterior distribution related to those of the conjugate prior? [2 marks]

Suppose that a rather devious type of mine contains an internal clock that ticks with period θ . Once the mine is armed it explodes on the next tick of the clock. Thus, the delay t between a random arming event and the following explosion is distributed:

$$t \sim \text{Uniform}(0, \theta).$$

You arm n mines and (from a safe distance) observe delays to the respective explosions of $t_1 \dots t_n$. You want to use these data to understand the properties of the internal clocks.

To begin, assume that all mines have clocks with exactly the same period θ .

- (c) What is the likelihood function for θ ? What is the maximum likelihood estimate of θ ? Do you think this is a reasonable estimate for small n (e.g. is it unbiased)? [2 marks]
- (d) Show that a conjugate prior for θ has the form (known as the Pareto distribution):

$$p(\theta) = \frac{1}{Z(\theta_0, k)} \Theta(\theta - \theta_0) \frac{1}{\theta^k}$$

where $\Theta(x)$ is the Heaviside function ($= 1$ for arguments > 0 and 0 otherwise), and θ_0 and k are the hyperparameters.

Calculate the normalising constant $Z(\theta_0, k)$. [3 marks]

- (e) Give the normalised posterior distribution on θ under this prior. What is the maximum a posteriori estimate? [2 marks]

Now you wonder whether it is indeed true that all the clocks share the same period. Assume you have access to reasonable values for θ_0 and k .

- (f) What is the marginal likelihood of your observed set of delays, under the model assumed thus far of a single θ ? [3 marks]
- (g) Suppose instead that each clock has a period drawn iid from the same Pareto prior. What is the marginal likelihood now? [3 marks]
- (h) If all the observed delays happened to be identical, which model would be favoured? You may work in the limit $n \gg 1$. [3 marks]

2.2 2019 Q1: Conjugacy in Regression [20 marks]

Also appears verbatim as 2022 Q1.

Consider linear regression for input-output pairs (x_n, y_n) , $n = 1 \dots N$ with weight vector w and output noise variance σ^2 :

$$y_n \sim \mathcal{N}(w^\top x_n, \sigma^2).$$

- (a) What is a conjugate prior? [2 marks]

- (b) State the form of the conjugate prior for w , assuming that σ^2 is known. You do not need to derive the result. [2 marks]

Now consider the case where σ^2 is unknown and is estimated along with w . We might choose an inverse-gamma prior on σ^2 :

$$p(\sigma^2) = \text{InvGamma}(\sigma^2; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} (\sigma^2)^{-\alpha-1} e^{-\beta/\sigma^2}.$$

- (c) Find the joint posterior on σ^2 and w under independent priors on w (of the form you identified above) and inverse-gamma on σ^2 . Your notation may be simpler if you introduce a matrix $X = [x_1 \dots x_N]$ and row vector $Y = [y_1 \dots y_N]$. Simplify your expression (discarding constants of proportionality) to isolate the functional dependence on σ^2 and w as far as possible. [6 marks]

Your result to the previous part should have shown that the independent prior defined above is *not* conjugate. The correct conjugate prior is a joint distribution with the form:

$$p(\sigma^2, w) = \text{InvGamma}(\sigma^2; \alpha, \beta) \mathcal{N}(w; \mu, \sigma^2 A^{-1}).$$

You may take $\mu = 0$.

- (d) Show that this is indeed a conjugate prior, and derive the updates to the parameters α , β , μ and A induced by the likelihood. [8 marks]
- (e) In lecture we considered Bayesian learning of parameters within A (for example, the diagonal entries) by maximising the evidence or marginal likelihood obtained by integrating out w alone, while maximising over σ^2 . Do you think the results for these parameters would have been different if we had integrated out both w and σ^2 ? You do not need to derive the result — just argue whether it will be the same or different based on your results here. [2 marks]

3 Topic 3: Gaussian Processes

3.1 2018 Q6: Periodic Gaussian Processes [20 marks]

Recall that a Gaussian Process (GP) prior over functions $f(t) \sim \mathcal{GP}(0, K)$ can (in certain cases) be derived from the Bayesian linear regression model

$$\beta \sim \mathcal{N}(0, I), \quad f(t) = \beta^\top \phi(t)$$

where $\phi(t) = [\phi_1(t), \phi_2(t), \dots]$ is a vector of nonlinear feature functions, and I is an identity matrix.

- (a) Derive the relationship between $\phi(t)$ and the GP covariance kernel $K(t, t')$. [4 marks]
- (b) Note that we have written the linear mapping above without noise. If we had incorporated noisy observations:

$$y \sim \mathcal{N}(\beta^\top \phi(t), \sigma^2)$$

what would be the (marginal) covariance matrix for a set of observations $\{y_i\}$ at inputs $\{t_i\}$? [2 marks]

The standard choice of kernel function for a Gaussian Process prior on periodic functions $f(t)$ is

$$K(t, t') = s^2 \exp\left(-\frac{2 \sin^2(\pi|t - t'|/p)}{l^2}\right).$$

- (c) Give interpretations of the hyperparameters s , p and l . [3 marks]
- (d) Sketch a draw from a GP with the product kernel

$$K(t, t') = s^2 \exp\left(-\frac{2 \sin^2(\pi|t - t'|/p)}{l^2}\right) \exp\left(-\frac{(t - t')^2}{2r^2}\right)$$

indicating the roles of s , p , l and r . [4 marks]

An alternative periodic kernel is simply $K(t, t') = s^2 \cos(2\pi(t - t')/p)$.

- (e) Show that this is a valid kernel by explicitly constructing a corresponding feature vector $\phi(t)$. [2 marks]
- (f) Based on this decomposition, describe the set of all functions that have non-zero density under the corresponding GP prior. [3 marks]
- (g) Why do you think the first form given above is the more commonly used? [2 marks]

3.2 2019 Q6: GPFA — Gaussian Processes for Unsupervised Learning [20 marks]

Consider a model for multivariate time series based on a set of latent functions of time $z_k(t)$, each drawn from an independent Gaussian process, which are then linearly combined with Gaussian observation noise to generate the observations:

$$\begin{aligned} z_k(\cdot) &\sim \mathcal{GP}(\mu_k(\cdot), \kappa_k(\cdot, \cdot)) & k = 1 \dots K \\ x_d(t) &\sim \mathcal{N}\left(\sum_k \Lambda_{dk} z_k(t), \psi_d\right) & d = 1 \dots D \end{aligned}$$

This model has sometimes been called Gaussian-process factor analysis or GPFA. Suppose we have observations of $x(t_n) = [x_1(t_n) \dots x_D(t_n)]^\top$ at a set of N times $t_n \in \{t_1 \dots t_N\}$. We collect these into a $D \times N$ matrix X , with $X_{dn} = x_d(t_n)$. Define also a $K \times N$ matrix Z of latent variables with $Z_{kn} = z_k(t_n)$.

- (a) Find the posterior distribution on Z . [3 marks]
- (b) Give the expectation-maximisation M-step needed to learn the maximum likelihood value of Λ , assuming that all the μ_k and κ_k are known. [3 marks]
- (c) Consider a new time t^* . Compute the predictive distribution $P(x(t^*)|X)$. [4 marks]
- (d) Suppose that we only observed a single output process at each time t_n . That is, rather than all of $x(t_n)$ we have only some $x_{d_n}(t_n)$. Assume that the identity of the particular dimension d_n that is observed does not depend on any latent z . Describe an approach to estimate the posterior on Z , maximum-likelihood $\hat{\Lambda}$ and predictive $P(x(t^*)|\mathcal{X})$ (where $\mathcal{X}$ is the set of observed values) in this case. [6 marks]
- (e) This model is related to the linear-Gaussian state-space model (LGSSM) that we considered in lecture. Under what parametric constraints (on both GPFA and the LGSSM) are the two equivalent? State conditions that are as general as possible. [4 marks]

3.3 2020 Q6: Tracking Drones [20 marks]

A drone is being flown along a linear aerial corridor, with camera traps set at locations $x_1 \dots x_L \in \mathbb{R}$. The l th trap reports a non-empty list of times $t_{l,1} \dots t_{l,n_l}$ at which the drone was at x_l , but the traps are not entirely reliable and so the lists may be incomplete.

We model the trajectory of the drone $x(t)$ as a draw from a Gaussian Process (GP) with smooth covariance function K .

- (a) Suppose $n_l = 1$ for all l and write $t = [t_{1,1} \dots t_{L,1}]^\top$. Give the predicted distribution for the location of the drone $x(t)$ at time t as a function of the data vector t . [2 marks]
- (b) How would your answer change for arbitrary n_l ? [2 marks]
- (c) Would your expression be valid if the traps were completely reliable? Explain. [4 marks]

Now consider estimating the drone's velocity $v(t)$. As differentiation is a linear operation, $x(t)$ and $v(t')$ are jointly Gaussian distributed for all t, t' and so form a two-output Gaussian process.

- (d) Find the covariance function $K_{xv}(t, t')$ between $x(t)$ and $v(t')$ in terms of the covariance function K on x . [4 marks]
- (e) Based on this result, what is the predicted distribution of $v(t)$ given the trap times? [4 marks]
- (f) Let $K(t, t') = e^{-(t-t')^2/(2\tau^2)}$. What is the covariance between $v(t)$ and $v(t')$ when $t - t' = \tau$? Is this a sensible prior? How might you construct a better one? [2 marks]

4 Topic 4: MLE / MAP Estimation

4.1 2017 Q2(b): MAP M-Step for HMM Transition Matrix [8 marks]

Consider a hidden Markov model (HMM) for observations $x_t \in \mathbb{R}^D$, $t = 1 \dots T$ with hidden states $s_t \in \{1 \dots K\}$, initial state distribution $P(s_1 = i) = \pi_i$, transition matrix $P(s_t = j | s_{t-1} = i) = \Phi_{ij}$, and output distributions $P(x_t | s_t = i) = A_i(x_t)$.

Given forward and backward messages:

$$\begin{aligned}\alpha_t(i) &= P(x_1, \dots, x_t, s_t = i | \Phi, \{A_j\}, \pi) \\ \beta_t(i) &= P(x_{t+1}, \dots, x_T | s_t = i, \Phi, \{A_j\})\end{aligned}$$

and a prior on the rows of Φ :

$$\Phi_{i,\cdot} \sim \text{Dirichlet}(\lambda), \quad \lambda \in \mathbb{R}_+^K,$$

derive the maximum a posteriori (MAP) M-step update for the transition matrix Φ by finding the analytic maximum of the free energy. [8 marks]

4.2 2020 Q2 / 2022 Q2: Optimisation — MLE for Markov Chains [20 marks]

Consider a Markov chain $X_1, X_2, \dots$ with states $1, \dots, K$ and transition matrix $P \in \mathbb{R}^{K \times K}$. The entry in row i and column j is

$$P_{ij} = \text{probability that } X_{t+1} = i \text{ given that } X_t = j,$$

so the sum of each column is 1. We observe the first n states $X_1 = x_1, \dots, X_n = x_n$ the chain visits, and denote by N_{ij} the transition counts

$$N_{ij} = \text{number of times that } x_{t+1} = i \text{ and } x_t = j \text{ for } t < n.$$

- (a) Suppose $f_1, \dots, f_K$ are convex functions and $g_1, \dots, g_K$ linear functions, all from $\mathbb{R}^d \rightarrow \mathbb{R}$. Show that the convex optimisation problem

$$\min \sum_{k \leq K} f_k(x_k) \quad \text{subject to } g_1(x_1) = 0, \dots, g_K(x_K) = 0 \quad (\text{A})$$

has the same solution(s) as are obtained by solving the K separate problems

$$\min f_k(x_k) \quad \text{subject to } g_k(x_k) = 0 \quad \text{for } k = 1, \dots, K. \quad (\text{B})$$

[3 marks]

- (b) Write out the log-likelihood of P as a function of the transition counts. Define a convex(!) optimisation problem with constraints whose solution is $\hat{P}$.

Hint: The logarithm is a concave function. If $\alpha, \alpha' \geq 0$ and h and h' are convex functions, then $\alpha h + \alpha' h'$ is also a convex function. [6 marks]

- (c) Consider the j th column $\hat{P}_j$ of $\hat{P}$. Specify a convex optimisation problem that has solution $\hat{P}_j$ and involves only the transition counts for the j column, $N_{1j}, \dots, N_{Kj}$. Show that

$$\hat{P}_{kj} = \frac{N_{kj}}{\sum_{i \leq K} N_{ij}}$$

for each $k \leq K$ solves this problem.

[8 marks]

Now suppose we are given a probability distribution π on the set $1, \dots, K$, and we change our model to

$$\mathcal{M}_\pi := \{\text{all Markov chains with states } 1, \dots, K \text{ and invariant distribution } \pi\}.$$

- (d) State maximum likelihood estimation in this model as a convex optimisation problem. Can (a) again be used to split the problem into K separate problems? [3 marks]

5 Topic 5: MCMC / Sampling

5.1 2022 Q4: Markov Chains and Sampling Algorithms [20 marks]

The first few questions below develop a (very primitive version of) a method known as the Propp-Wilson algorithm. The final question is a separate problem.

Consider a finite set $\mathcal{K} = \{1, \dots, K\}$. Let f_θ be a function $\mathcal{K} \rightarrow \mathcal{K}$ parametrised by some value θ . We generate parameter values $\Theta_1, \dots, \Theta_n$ at random, and define

$$F_n(x) := f_{\Theta_1} \circ \dots \circ f_{\Theta_n}(x) = f_{\Theta_1}(f_{\Theta_2}(\dots f_{\Theta_n}(x) \dots))$$

and reversely

$$G_n(x) := f_{\Theta_n} \circ \dots \circ f_{\Theta_1}(x).$$

- (a) We fix any x , and generate $\Theta_1, \dots, \Theta_n$ either i.i.d., or from a Markov chain, or from a hidden Markov model. In which of these cases is the sequence $(G_1(x), G_2(x), \dots, G_n(x))$ a Markov chain? [3 marks]
- (b) Now assume $\Theta_1, \dots, \Theta_n$ are i.i.d. For some fixed x , do $F_n(x)$ and $G_n(x)$ have the same distribution? Does the sequence $(F_1(x), F_2(x), \dots)$ have the same distribution as $(G_1(x), G_2(x), \dots)$? [3 marks]

Suppose we can find functions f_θ and a distribution for the Θ_i such that F_n always becomes a constant function eventually, that is, there are some random values N in $\mathbb{N}$ and X in $\mathcal{K}$ such that $F_N(x) = X$ for all initial values $x \in \mathcal{K}$.

- (c) Let π be the distribution of X . What is the distribution of $F_{N+1}(x)$? [3 marks]
- (d) Suppose you have access to the random values $\Theta_1, \Theta_2, \dots$. Specify an algorithm, as a single while loop, that takes input x and outputs a single draw from π . [3 marks]
- (e) Which method would be preferable to generate a single draw from π : The algorithm in (d), or a Metropolis-Hastings algorithm with invariant distribution π ? [3 marks]
- (f) We want to sample from a distribution with a very simple “staircase” density p . We divide the area under p into three blocks, and define a Markov chain that generates samples $x_1, x_2, \dots$ as follows. Start with any $X_1 \in [a, b]$. For each $i > 1$:
- Draw $T_i \sim \text{Uniform}[0, p(X_{i-1})]$.
 - Select the block k which contains (X_{i-1}, T_i) .
 - Sample a point (X_i, Y_i) uniformly from the area in block k .
 - Discard the vertical coordinate Y_i and keep X_i as the i th sample.

Is this a valid sampler for p ? More precisely: If we assume that the previous sample X_{i-1} is distributed according to p , is X_i marginally distributed according to p ? [5 marks]

6 Topic 6: Model Comparison / Marginal Likelihood

Covered within the Conjugate Mines question (2018/2020 Q1, parts f–h). See Section 2.1.

7 Topic 7: Mixture Models and EM

7.1 2019 Q2: A Mixture of Linear Experts [20 marks]

A mixture of experts is a latent variable model for supervised learning. The “experts” are each simple probabilistic supervised learning models. Here, we consider the case where each expert corresponds to vector-output linear regression with Gaussian noise.

We are given input-output vector pairs (x_n, y_n) , $n = 1 \dots N$. For each pair, we introduce a discrete latent “gating” variable $s_n \in \{1 \dots K\}$, with the probability distribution for s_n determined by the distance of the corresponding input x_n from each of K points μ_k . The k th linear expert has weight matrix W_k , but all share the same diagonal output noise covariance Ψ .

The joint model can be written:

$$P(s_n = k | x_n, \Theta) = \pi_k(x_n) = \frac{e^{-\frac{1}{2}\|x_n - \mu_k\|^2}}{\sum_{k'=1}^K e^{-\frac{1}{2}\|x_n - \mu_{k'}\|^2}}$$
$$P(y_n | s_n = k, x_n, \Theta) = \mathcal{N}(W_k x_n, \Psi)$$

where the parameters are $\Theta = \{\mu_1 \dots \mu_K, W_1 \dots W_K, \Psi\}$.

We will derive an Expectation Maximisation (EM) algorithm to find the maximum-likelihood parameters of this model.

- (a) Show that $P(y_n, s_n | x_n, \Theta)$ is a joint exponential family model by expressing it in the standard exponential family form, identifying the sufficient statistic functions, and giving the natural parameters as functions of x_n and the conventional parameters defined above. [6 marks]
- (b) Find the posterior distribution $P(s_n = k | x_n, y_n, \Theta)$. [4 marks]
- (c) Writing r_{nk} for the posterior probability defined in part (b), derive closed-form M-step updates for:
 - W_k [4 marks]
 - μ_k [6 marks]

8 Topic 8: Nonlinear Regression / ARD (GP Connection)

8.1 2017 Q6: Nonlinear Regression [20 marks]

Consider the Bayesian nonlinear regression model:

$$\begin{aligned} y_i &= \beta^\top \phi(x_i) + \epsilon, & \epsilon &\sim \mathcal{N}(0, \sigma^2) \\ \beta &\sim \mathcal{N}(0, A^{-1}), & A &= \text{diag}[\alpha_1 \dots \alpha_K] \end{aligned}$$

where $\phi : \mathbb{R}^D \rightarrow \mathbb{R}^K$ is a vector-valued function from the D -dimensional input space of x to a K -dimensional feature space, and β is a K -dimensional weight vector. It will be useful to collect the vectors $\phi(x_i)$ for points $\{x_i\}_{i=1}^N$ into a $(K \times N)$ -dimensional matrix Φ , and the corresponding values y_i into a $(1 \times N)$ row vector Y .

- (a) Write down the joint distribution $P(Y|\Phi, \beta, \sigma^2)$ as a vector normal on $Y^\top$. [3 marks]
- (b) Find the posterior distribution on β given the N observed (x_i, y_i) pairs. [5 marks]
- (c) Find the marginal likelihood $P(Y^\top|\Phi, A, \sigma^2)$ by first arguing that it is normal, and then computing the mean and variance using the standard properties of moments of conditional distributions. [3 marks]
- (d) Consider optimising the marginal likelihood with respect to the hyperparameters $\{\alpha_i\}$. Reasoning by analogy to automatic relevance determination (ARD) for linear regression, describe the form of the result you expect to obtain after optimisation. How would your interpretation of this result differ in the context of this regression model (rather than linear regression)? [3 marks]
- (e) Differentiating the marginal likelihood directly is possible, but not necessarily the most efficient method of optimisation. One alternative is to treat β as a latent variable, and use expectation maximisation (EM). Compute the M-step update for A in this approach. [3 marks]

An alternative parameterisation would be to fix the prior on β to $\mathcal{N}(0, I)$, and instead introduce a D -dimensional hyperparameter γ to the regression model:

$$y_i = \beta^\top \phi(\gamma^{-1} \circ x_i) + \epsilon$$

where $\circ$ is the Hadamard or elementwise product. Again, it is possible to optimise the marginal likelihood with respect to γ . This approach has the advantage of being compatible with Gaussian process (GP) regression, where the product $\phi(x_i)^\top \phi(x_j)$ is replaced by a covariance kernel $K(x_i, x_j)$.

- (f) Consider using the exponentiated quadratic kernel

$$K(x_i, x_j) = e^{-\|x_i - x_j\|^2}.$$

Interpret the alternative hyperparameters γ within this regression model. If they behave in the same way as the α s of ARD, what can be concluded about the regression result? [3 marks]